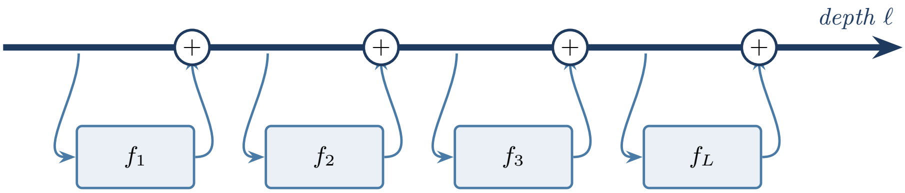
“SGD is a ResNet too — the residual stream is the weights”

One update rule, three different axes to rotate it onto.

A Transformer / ResNet

— residual stream over **depth**



attention / MLP blocks only *perturb* the stream

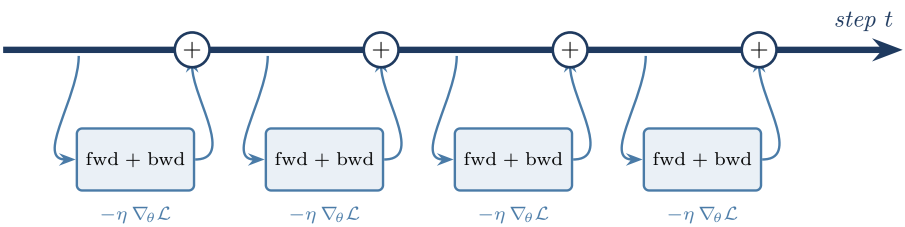
$$h_{\ell+1} = h_\ell + f_\ell(h_\ell)$$

fixed unit coefficients

identical

B SGD

— the **weights** are the residual stream, over **optimizer steps**



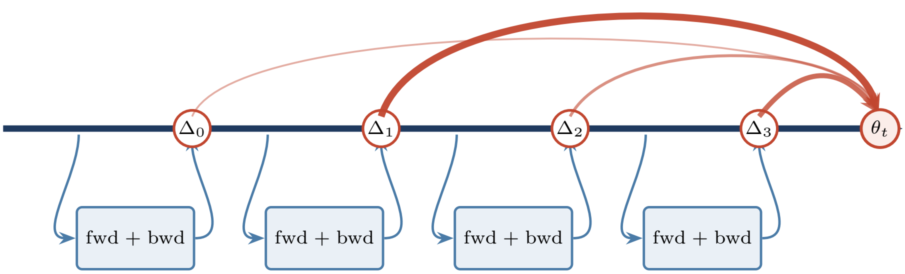
$$\theta_{t+1} = \theta_t - \eta\, g_t$$

same shape as (A)

C ...so take “Attention Is All You Need” literally

Kimi’s AttnRes does exactly this to axis (A). Karpathy’s needling question: so why not to axis (B)?

softmax attention over the *history of updates*



$$\theta_t = \sum_{i<t} \alpha_{i\to t}\,\Delta_i, \qquad \alpha_{i\to t} = \mathrm{softmax}_i\big(\phi(q_t, k_i)\big)$$

Momentum / Adam already do this with *fixed, geometric* $\alpha_{i\to t} = (1-\beta)\beta^{\,t-i}$ — i.e. **linear** attention over the step axis. The open question is the softmax, content-dependent version.

The same operator, rotated onto a different axis

axis	fixed accumulation	attention version
token j	(an RNN)	self-attention *(Vaswani et al.)*
depth ℓ	residual stream $h_\ell = h_{\ell-1} + f(h_{\ell-1})$	AttnRes *(Kimi, 2603.15031)*
step t	SGD / momentum $\theta_t = \theta_{t-1} - \eta g_t$	**?** *(Karpathy)*